Maximum Ambiguity of Ordered Adjacent-Difference Records

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Abstract

An ordered adjacent-difference record retains the absolute differences of consecutive entries of a permutation. Its ambiguity is the number of permutations giving that exact word. We study its maximum $M(n)$ without asserting an unsupported exact formula. We prove a signed-walk characterization, endpoint-sensitive gluing, and existence of the global exponential growth rate, with a lower bound greater than 1.28142. We completely classify the infinite family of two unit-step blocks separated by one jump, and derive constant-run and fixed-endpoint antichain bounds. Examples refute simple structural heuristics, including the necessity of a unit step in every optimizer. The underlying reconstruction mapping has prior literature; maximum fiber structure remains the central open problem.

Preprint; not externally peer reviewed. AI use and verification are described at the end.

Introduction and research strategy

The exploratory route began with permutations as completely enumerable objects and then deliberately discarded information: retain each adjacent distance, but erase the starting label and every step direction. Grouping permutations by the information left behind leads naturally to a fiber-size question. The extreme fiber was not selected in advance for its appearance; it emerged from grouping all small permutations and comparing group sizes.

The data suggested two different questions. General orders displayed several competing shapes and irregular maximum values, whereas powers of two displayed one stable recursive pattern. The former is the open extremal problem in this paper; the latter is isolated in the companion dyadic conjecture. Subsequent literature checks established that the difference mapping and its reconstruction problem were already known. The formation narrative concerns how this project arrived at the maximum-fiber question, not historical invention of the map.

Research strategy. First convert fibers into signed self-avoiding walks, then retain endpoint information for gluing. To make the structural discussion concrete, we also solve an infinite local family consisting of two unit-step blocks separated by one jump. Repeated distances impose forced directions, while fixed endpoints give an antichain bound. These results illustrate what a small structural invariant can accomplish, and why it still does not classify unrestricted optimizers. The exponential growth theorem uses gluing with fixed endpoints and does not require an exact formula for $M(n)$.

1 The extremal problem

For $\pi \in S_n$, define the ordered word

$$\Delta(\pi) = (|\pi_2 - \pi_1|, \ldots, |\pi_n - \pi_{n-1}|).$$

For a word $d \in \{1, \dots, n-1\}^{n-1}$, set

$$A_n(d) = \#\{\pi \in S_n : \Delta(\pi) = d\}, \qquad M(n) = \max_d A_n(d).$$

Take $M(1) = 1$ for the empty word. We count labeled oriented permutations: neither complementation nor reversal is quotiented out. Reordering the entries of d generally changes its fiber.

Problem 1.1. Determine $M(n)$ and classify every maximizing word. Determine its exponential growth rate and the structure of words whose ambiguity is close to maximum.

No defensible general-n formula is currently supplied, so this is an extremal problem rather than a conjecture asserting such a formula. The separate dyadic conjecture concerns only powers of two.

2 Prior work and scope

The mapping Δ is not new. De Biasi proves NP-completeness of deciding whether a prescribed ordered difference word has a permutation realization [1]. Bulteau–Giraudo–Vialette use the same adjacent absolute differences in their study of additive disorder, which optimizes the *sum* of those differences [2]. That objective differs from maximizing the number of permutations with one fixed entire word. Earlier public questions also concern realizability and equal-distance traversals with prescribed endpoints [3, 4].

Focused searches through 14 November 2026 did not locate a general solution to Problem 1.1 in the checked sources. This is a limited literature finding, not certification that no such result exists. In particular, the established reconstruction problem must not be advertised as a newly invented object. Its worst-case NP-completeness also does not prove that the present extremal problem, or any fixed order such as 32, has no short structural proof.

3 Signed walks and symmetries

Proposition 3.1. For a fixed word d, $A_n(d)$ equals the number of sign words $\varepsilon \in \{-1, 1\}^{n-1}$ whose partial sums

$$s_0 = 0, \qquad s_j = \sum_{i=1}^{j} \varepsilon_i d_i$$

are distinct and satisfy $\max_j s_j - \min_j s_j = n - 1$. Thus $A_n(d) \le 2^{n-1}$.

Proof. The partial sums of a realizing permutation are its entries translated by $-\pi_1$, so they have the stated properties. Conversely, n distinct integers in an interval of width $n-1$ fill that interval. Translation by $1 - \min s_j$ is therefore the unique translation producing $[n]$. Since every $d_i > 0$, the signs are recoverable from the permutation. This proves the bijection and the bound. □

Complementation $\pi_i \mapsto n + 1 - \pi_i$ is a fixed-point-free involution within each fiber for $n \ge 2$, so every nonzero fiber size is even. Reversal gives $A_n(d) = A_n(\operatorname{rev} d)$.

Let $e_d(i)$ count realizations ending at i. Complementation implies $e_d(i) = e_d(n+1-i)$. Unlike the scalar $A_n(d)$, this vector retains information needed for gluing.

Proposition 3.2. For any realizable word d on n labels, the word $(2d, 1, 2\operatorname{rev} d)$ on $2n$ labels has ambiguity

$$Q(d) = 2\sum_{i=1}^{n} e_d(i)^2 + 2\sum_{i=1}^{n-1} e_d(i)e_d(i+1) \geq \frac{2A_n(d)^2}{n}.$$

Consequently $M(2n) \geq 2M(n)^2/n$.

Proof. All differences in either block are even, so each block exhausts one parity class. An odd label $2i-1$ and an even label $2j$ have distance one precisely when $j = i$ or $j = i-1$. Summing the compatible endpoint counts for both parity orders gives the displayed quadratic form. Its inequality follows from $\sum e_i^2 \geq (\sum e_i)^2/n$ and nonnegativity of adjacent products. □

4 Existence of a global growth constant

Let $B(n)$ be the largest fiber size with both first and last labels prescribed, maximizing over the word and the two labels. Set $B(1) = 1$.

Theorem 4.1. There exists $\gamma \in [1, 2]$ such that

$$\lim_{n\to\infty} M(n)^{1/n} = \gamma, \qquad \gamma = \sup_{n\geq 1}\left(\frac{M(n)}{n}\right)^{1/n}.$$

Proof. Concatenate a fixed-endpoint family on $[n]$ with a fixed-endpoint family on $\{n+1, \ldots, n+m\}$. Both internal words and the connecting distance are fixed, as are the two outer endpoints. Distinct pairs give distinct permutations. Therefore

$$B(n+m) \geq B(n)B(m), \qquad B(n) \leq M(n) \leq n(n-1)B(n) \quad (n \geq 2).$$

For completeness, if $a_n = \log B(n)$, then $a_{n+m} \geq a_n + a_m$. Fix k and write $n = qk + r$, $0 \leq r < k$. If $r = 0$, superadditivity gives $a_n \geq qa_k$ directly; if $r > 0$, it gives $a_n \geq qa_k + a_r \geq qa_k$. Thus $\liminf a_n/n \geq a_k/k$. Taking the supremum in k proves convergence of a_n/n to its supremum. The polynomial comparison transfers this limit to $\log M(n)/n$, and Proposition 3.1 bounds it by $\log 2$.

Put $H(n) = M(n)/n$. Proposition 3.2 says $H(2n) \geq H(n)^2$. Iterating and taking $(2^j n)$th roots gives $\gamma \geq H(n)^{1/n}$ for every n. Conversely $H(n)^{1/n} \to \gamma$, so the supremum cannot be smaller than γ. □

Define D_1 to be empty and $D_{2n} = (2D_n, 1, 2D_n)$. Exact endpoint-matrix recurrence, proved in the companion dyadic note, gives

$$A_{128}(D_{128}) = 7794829488660802.$$

Since $M(128)$ is at least this witnessed count, Theorem 4.1 yields

$$\gamma \geq \left(\frac{7794829488660802}{128}\right)^{1/128} = 1.28142195708018\ldots.$$

This is a lower bound on the global growth constant. An upper bound for the growth rate of the dyadic construction alone would not be an upper bound on γ.

5 Finite data and endpoint obstructions

n	1	2	3	4	5	6	7	8
$M(n)$	1	2	2	6	4	8	8	30
Maximizers	1	1	3	1	6	3	5	1
n	9	10	11	12	13	14	15	16
$M(n)$	12	24	24	68	40	72	72	430
Maximizers	20	5	6	1	2	2	13	1

The first row range $n \le 8$ was freshly re-enumerated over all permutations. The range $9 \le n \le 16$ is an archived completed exact branch-and-bound result, not a fresh exhaustive run. Every listed maximizing witness through 16 was freshly recounted by an independent signed-walk procedure. Recounting witnesses certifies their counts, not the absence of unlisted competitors. The archived catalogue and the new checks are supplied separately in `research/archive/catalogue.json` and `permutation_verification.json`.

The values $M(8) = 30 > M(9) = 12$ exclude monotonicity. More subtly, the word $u = (4, 2, 4, 1, 4, 2)$ on seven labels has only six realizations, all ending at 4, while $M(7) = 8$. Nevertheless gluing u with its reversal as in Proposition 3.2 yields 72 realizations on 14 labels, attaining the archived $M(14)$. Thus discarding locally nonmaximal words can discard globally optimal building blocks.

6 Local structure: forced directions and a solvable family

Proposition 6.1. Let $R(d)$ be the number of maximal constant runs in a distance word. Then

$$A_n(d) \le 2^{R(d)}.$$

If a run has length ℓ and distance a, realizability requires $\ell a \le n - 1$. It also requires $\gcd(d_1, \dots, d_{n-1}) = 1$ for $n \ge 2$.

Proof. Opposite signs on two consecutive equal distances return immediately to a previously visited label. Thus every constant run has one direction, and at most one sign choice per run remains. The translation is unique by Proposition 3.1, giving the first bound. A run spans ℓa within $[n]$, proving the second assertion. If all distances share a divisor $g > 1$, every visited label has the same residue modulo g, whereas $[n]$ does not. □

These are necessary conditions, not a reconstruction criterion. For example $(1, 1, 2)$ at order four has gcd one and obeys the run-length bounds, but is not realizable: the first three entries form a monotone interval of length three, and the remaining label is at distance either one or three from its last endpoint. A constant word of ones has exactly two realizations, while any other positive constant word is unrealizable. Thus a large number of repeated short steps creates forced motion rather than large ambiguity.

An infinite family permits a complete answer. For $a, b \ge 2$, $a + b = n$, put

$$w_{a,b,t} = (\underbrace{1, \dots, 1}_{a-1}, t, \underbrace{1, \dots, 1}_{b-1}), \qquad 1 \le t < n.$$

Theorem 6.2. The exact fiber size is

$$A_n(w_{a,b,t}) = 2\big(\mathbf{1}_{t=1} + \mathbf{1}_{t=a} + \mathbf{1}_{t=b} + \mathbf{1}_{t=n-1}\big).$$

In particular it is at most four, with equality precisely when $a = b = t$.

Proof. Each unit-step block must be monotone, hence visits an interval. Two disjoint intervals covering $[n]$ must be its lower and upper intervals. If the first block is $[1, a]$ and the second is $[a+1, n]$, the four independent orientations give bridge lengths $1, b, a, n-1$, respectively. If the first block is the upper interval of length a, the corresponding four lengths are $n-1, a, b, 1$. Since $a, b \geq 2$, the two orientations of each block are distinct. All eight resulting permutations are distinct, and this list exhausts the possible interval partitions. Counting repeated bridge values proves the formula. Only $a = b$ can make two of its four indicated values coincide. □

For example, $(1, 2, 1)$ has four realizations at order four, whereas the unrestricted optimizer $(2, 1, 2)$ has six. For every $m \geq 2$, the word $(1^{m-1}, m, 1^{m-1})$ has exactly four realizations at order $2m$. Consequently, simply lengthening two monotone blocks does not yield exponential ambiguity. The construction is easy to understand precisely because its interval states are too rigid to capture general optimizers.

7 A fixed-endpoint bound from positive subset sums

Let $H_d(s, t)$ count the realizations of d starting at s and ending at t, and let $L = \sum_i d_i$.

Proposition 7.1. For $n \geq 2$,

$$H_d(s, t) \leq \binom{n-1}{\lfloor (n-1)/2 \rfloor}.$$

Moreover $H_d(s, t) = 0$ unless $L + t - s$ is an even integer between 0 and $2L$.

Proof. The set P of positive signed steps must satisfy

$$\sum_{i \in P} d_i = \frac{L + t - s}{2}.$$

Since all d_i are positive, distinct subsets with this sum form an antichain: none properly contains another. To bound an antichain on $m = n - 1$ positions, expose a random ordering of the positions and its chain of initial subsets. A fixed size-k subset appears with probability $1/\binom{m}{k}$, and at most one antichain member appears on any chain. Therefore

$$\sum_{P} \binom{m}{|P|}^{-1} \leq 1,$$

implying the central-binomial bound. Every endpoint-fixed realization determines one such subset; additional collision and range restrictions can only reduce their number. The parity and range condition follows from the displayed subset-sum identity. □

This is a standard antichain argument applied to the present encoding, not a sharp global fiber theorem. It retains endpoint and total-length information, but discards the order of collisions. That loss explains why it does not approach the dyadic candidate counts closely enough to prove their optimality.

There is another useful warning about structural guesses. The archived order-13 maximum is 40, attained by

$$(4, 2, 4, 3, 2, 9, 2, 4, 3, 4, 2, 4),$$

which contains no distance one. Its count of 40 was freshly verified by signed walks. Thus the claim that every maximizing word must contain a unit step is inconsistent with the archived complete classification. The new checks in `verify_extensions_v2.py` also verify the two-interval formula in 1001 cases through order 16 and the necessary run and antichain bounds on all permutations through order 8. Neither those checks nor the local theorems replace the unrestricted extremal problem.

8 A valid search bound and an open structural task

For a set S of unused labels and current label x, define recursively

$$U(\varnothing, x) = 1, \qquad U(S, x) = \max_{d} \sum_{\substack{y \in S \\ |y-x|=d}} U(S \setminus \{y\}, y).$$

This bounds the number of continuations with any one fixed remaining word: its first distance selects one summand group, and induction bounds every suffix. At a prefix shared by many partial paths, add the bounds for all current states with their multiplicities. Every common suffix is bounded by this sum, so pruning below the incumbent is valid; pruning equality would lose tied maximizers.

This bound supports exact small searches but still has exponentially many unused-label states. A theoretical advance would replace the full state by a much smaller invariant while controlling endpoint concentration and collisions. The signed-walk and gluing results do not yet provide that invariant, an exact general formula, or a characterization of near-maximal words.

Provenance and use of AI

This is a research draft, not a peer-reviewed publication. The problem arose in an AI-assisted exploratory project. AI tools were used substantially for literature searches, mathematical derivations, verification programs, and drafting. The proofs and bounded checks are supplied for independent scrutiny; neither an exhaustive novelty claim nor a general proof beyond the stated results is asserted.

References

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